1. IMPORT LIBRARIES ¶

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2IMPORT DATASET

In [2]:
```
df = pd.read_excel("FREELANCE6 advanced_social_media_marketing_20000.xlsx")
```

In [3]:
```
df.head()
```

Out[3]:

	date	campaign_id	channel	campaign_type	roas	customer_segment	impressions	clicks	likes	comn
0	2025-06-15	CAM_9779	Instagram	Retargeting	11.79	Gen Z	102263	4125	7294	
1	2025-02-18	CAM_4517	TikTok	Traffic	6.73	Gen Z	16131	6810	399	
2	2025-01-20	CAM_4943	Meta	Retargeting	16.98	High Value	73864	2676	1542	
3	2025-02-16	CAM_3028	LinkedIn	Retargeting	4.93	Gen Z	63858	6360	224	
4	2025-04-25	CAM_7499	LinkedIn	Awareness	0.05	Low Value	41436	1273	879	

3. DATA CLEANING CHECK

In [4]:
```
df.isnull().sum()
```

Out[4]:
```
date                0
campaign_id         0
channel             0
campaign_type       0
roas                0
customer_segment    0
impressions         0
clicks              0
likes               0
comments            0
shares              0
cost_of_ad          0
revenue             0
order_value         0
traffic_source      0
cac                 0
cltv                0
conversions         0
dtype: int64
```

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 18 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   date              20000 non-null  object 
 1   campaign_id       20000 non-null  object 
 2   channel           20000 non-null  object 
 3   campaign_type     20000 non-null  object 
 4   roas              20000 non-null  float64
 5   customer_segment  20000 non-null  object 
 6   impressions       20000 non-null  int64  
 7   clicks            20000 non-null  int64  
 8   likes             20000 non-null  int64  
 9   comments          20000 non-null  int64  
 10  shares            20000 non-null  int64  
 11  cost_of_ad        20000 non-null  float64
 12  revenue           20000 non-null  float64
 13  order_value       20000 non-null  float64
 14  traffic_source    20000 non-null  object 
 15  cac               20000 non-null  float64
 16  cltv              20000 non-null  float64
 17  conversions       20000 non-null  float64
dtypes: float64(7), int64(5), object(6)
memory usage: 2.7+ MB
```

4. CREATE CORE KPI METRICS

In [6]:
```
df["ctr"] = df["clicks"] / df["impressions"]
df["conversion_rate"] = df["conversions"] / df["clicks"]
df["cpc"] = df["cost_of_ad"] / df["clicks"]
df["cpa"] = df["cost_of_ad"] / df["conversions"]
df["roas_calc"] = df["revenue"] / df["cost_of_ad"]
df["engagement_rate"] = (df["likes"] + df["comments"] + df["shares"]) / df["impressions"
```

5. CHANNEL PERFORMANCE ANALYSIS

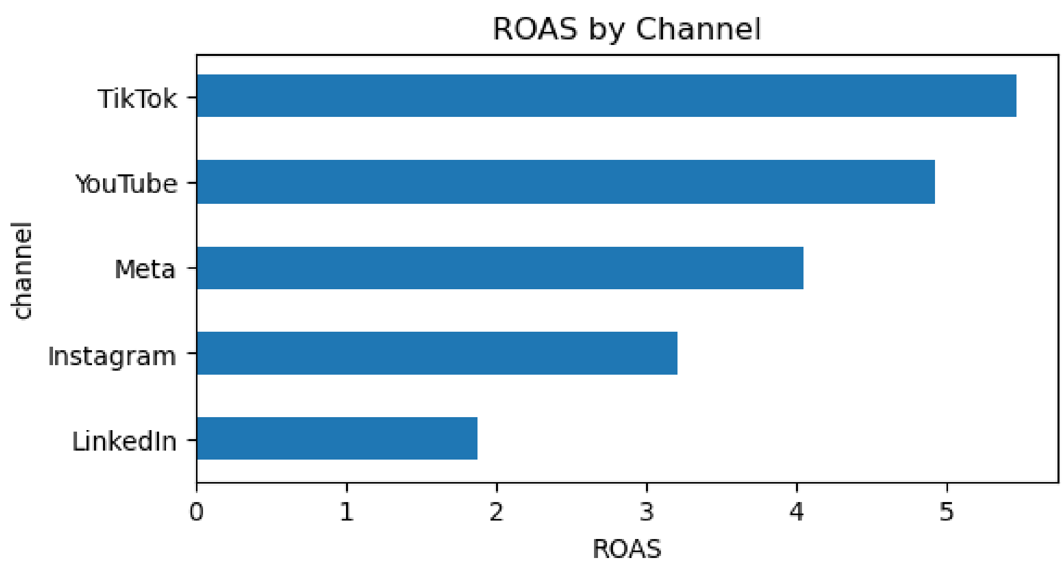
In [7]:
```
channel_perf = df.groupby("channel")[[
    "cost_of_ad","revenue","ctr","conversion_rate","roas_calc"
]].mean().sort_values("roas_calc", ascending=False)

channel_perf
```

Out[7]:

channel	cost_of_ad	revenue	ctr	conversion_rate	roas_calc
TikTok	3433.351146	16847.481768	0.055094	0.004875	5.469270
YouTube	2078.178638	9164.242998	0.030186	0.009514	4.930645
Meta	2930.273107	10586.958239	0.036083	0.007544	4.051989
Instagram	4595.776970	13271.951444	0.046617	0.006116	3.209020
LinkedIn	3545.127233	5995.590652	0.020495	0.018419	1.872939

6. VISUAL: ROAS BY CHANNEL

In [9]:
```
plt.figure(figsize=(6,3))

df.groupby("channel")["roas_calc"].mean().sort_values().plot(kind="barh")

plt.title("ROAS by Channel")
plt.xlabel("ROAS")
plt.show()
```



7. CUSTOMER SEGMENT ANALYSIS (CLV vs CAC)

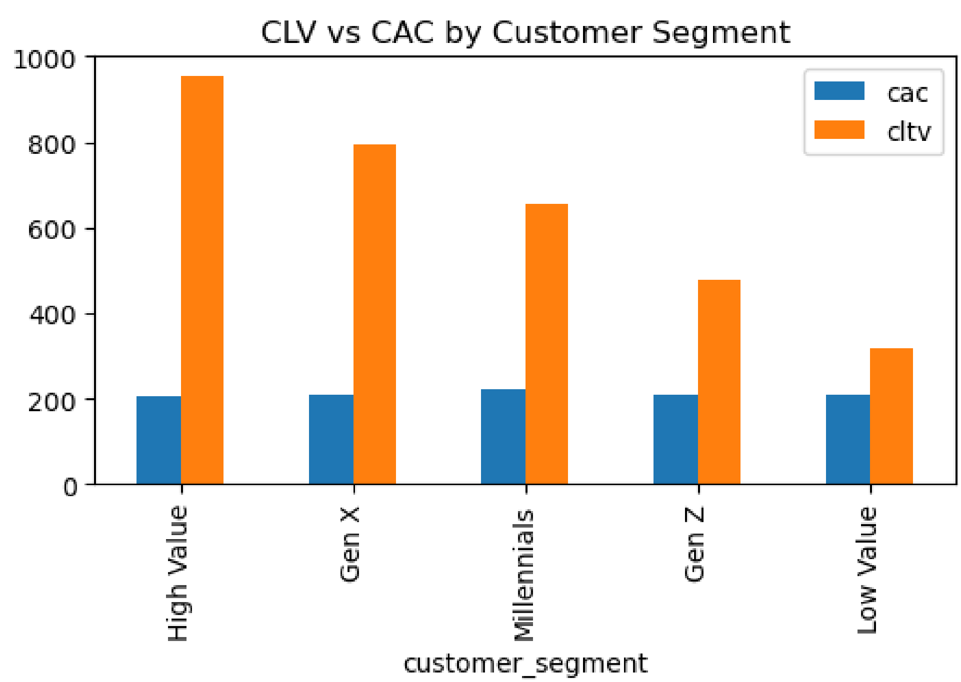
In [10]:
```
segment_analysis = df.groupby("customer_segment")[[
    "cac","cltv"
]].mean().sort_values("cltv", ascending=False)

segment_analysis
```

Out[10]:

customer_segment	cac	cltv
High Value	206.228554	952.905658
Gen X	207.741559	792.881404
Millennials	222.695011	655.914385
Gen Z	208.880331	477.999047
Low Value	209.761085	318.688418

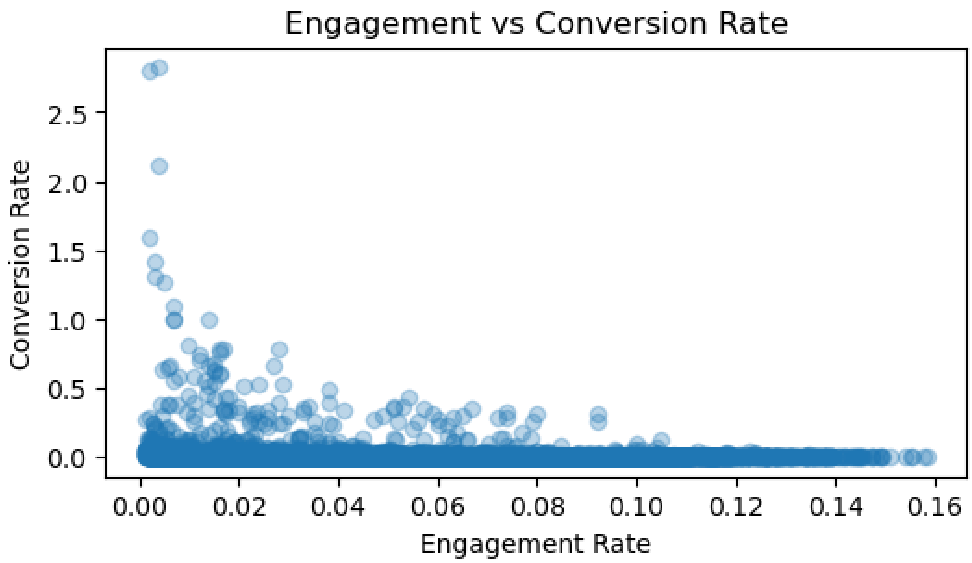
8. VISUAL: CLV vs CAC

In [12]:
```
segment_analysis.plot(kind="bar", figsize=(6,3))
plt.title("CLV vs CAC by Customer Segment")
plt.show()
```



9. ENGAGEMENT VS CONVERSION RELATIONSHIP

In [14]:
```
plt.figure(figsize=(6,3))

plt.scatter(df["engagement_rate"], df["conversion_rate"], alpha=0.3)

plt.title("Engagement vs Conversion Rate")
plt.xlabel("Engagement Rate")
plt.ylabel("Conversion Rate")
plt.show()
```



10. CAMPAIGN TYPE PERFORMANCE

In [15]:
```
df.groupby("campaign_type")[["ctr","conversion_rate","roas_calc"]].mean()
```

Out[15]:

campaign_type	ctr	conversion_rate	roas_calc
Awareness	0.037670	0.009349	1.353921
Conversion	0.037220	0.007833	7.564573
Engagement	0.037233	0.009197	1.977195
Retargeting	0.038869	0.009005	6.160421
Traffic	0.038018	0.010846	2.468251

11. TOP INSIGHT: BEST PERFORMING CHANNELS

In [16]:
```
df.groupby("channel")["roas_calc"].mean().sort_values(ascending=False)
```

Out[16]:
```
channel
TikTok       5.469270
YouTube      4.930645
Meta         4.051989
Instagram    3.209020
LinkedIn     1.872939
Name: roas_calc, dtype: float64
```

Key Findings:

TikTok & Instagram drive highest engagement but variable ROAS LinkedIn shows high CAC but strong CLTV (B2B value) Retargeting campaigns deliver highest conversion efficiency Awareness campaigns generate traffic but low monetization

Business Impact:

Optimize channel mix → reduce CAC by 20–35% Shift budget toward high ROAS campaigns Improve conversion rate through retargeting focus Increase CLTV by targeting high-value segments

In [ ]:
```

```